17/10/2025

F.T.

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

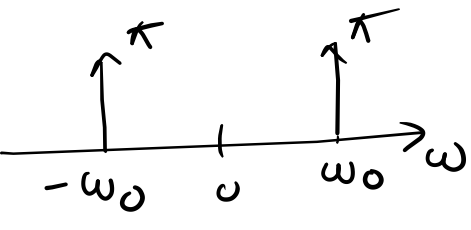
$\delta(t) \rightleftharpoons 1$

$1 \rightleftharpoons 2\pi\delta(\omega)$

$e^{j\omega_0 t} \rightleftharpoons 2\pi\delta(\omega-\omega_0)$

Ex

$$x(t) = \cos\omega_0 t = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$



$$e^{j\omega_0 t} \rightleftharpoons \frac{2\pi\,\delta(\omega-\omega_0)}{2}$$

$$e^{-j\omega_0 t} \rightleftharpoons \frac{2\pi\,\delta(\omega+\omega_0)}{2}$$

$$\frac{e}{2}$$

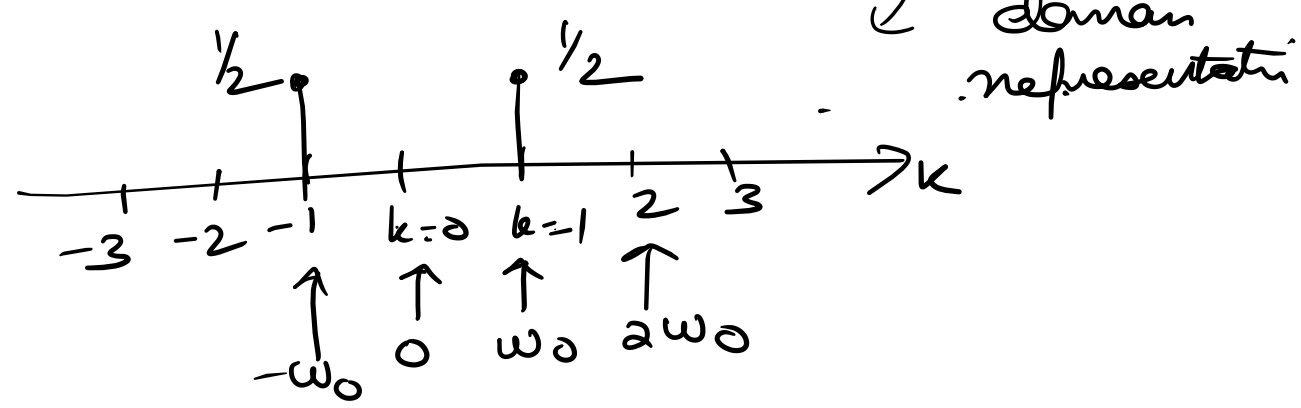
$$x(t) = \cos\omega_0 t \rightleftharpoons \pi\,\delta(\omega-\omega_0) + \pi\,\delta(\omega+\omega_0)$$

$\cos \omega_0 t \equiv$ periodic signal (C.T.)

$\cos \omega_0 t \xrightarrow{F.S.} a_1 = \frac{1}{2},\ a_{-1} = \frac{1}{2}$ & rest of all a_k's are zero.

$\Downarrow$

discrete line spectrum



for F.S., in the freq. domain, the freq. axis is discretized.

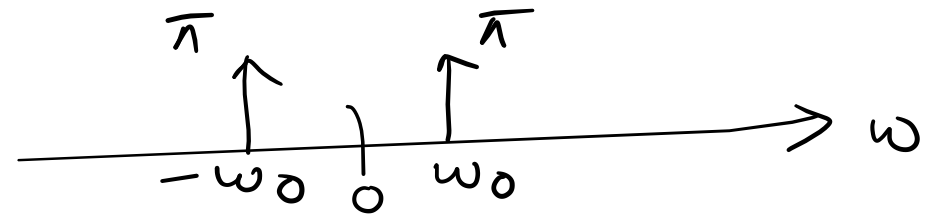
$\cos \omega_0 t \equiv$ C.T. periodic signal

$x(t) * \delta(t) = x(t)$

$\delta(t) = 0 \quad t \neq 0$

$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$

$\cos \omega_0 t \xrightleftharpoons[IF.T]{F.T}$



EX

C.T. periodic signal

$$x(t) \xrightarrow{F.S.} \{a_k\}$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Compute F.T.

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \rightleftharpoons \sum_{k=-\infty}^{\infty} a_k 2\pi \delta(\omega - k\omega_0)$$

$1 \rightleftharpoons 2\pi\delta(\omega)$

$e^{jk\omega_0 t} \rightleftharpoons 2\pi\delta(\omega - k\omega_0)$

$\delta(t) \rightleftharpoons 1$

EX



$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\frac{2\pi}{T}t} dt$$

$$= \frac{1}{T} = a_0$$

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{jk\omega_0 t}$$

$$X(j\omega) \rightleftharpoons x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

$$= \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$

Ex $\quad x(t) = \frac{2}{1+t^2}$

$x(t) \rightleftharpoons X(j\omega)$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$e^{-a|t|} \rightleftharpoons \frac{2a}{a^2+\omega^2}$

$e^{-|t|} \rightleftharpoons \frac{2}{1+\omega^2}$

$$\frac{2}{1+\omega^2} = \int_{-\infty}^{\infty} e^{-|t|}\, e^{-j\omega t}\, dt$$

change ω with $-\omega$.

$$\frac{2}{1+\omega^2} = \int_{-\infty}^{\infty} e^{-|t|}\, e^{j\omega t}\, dt$$

Swap t & ω

$$\frac{2}{1+t^2} = \int_{-\infty}^{\infty} e^{-|\omega|}\, e^{j\omega t}\, d\omega$$

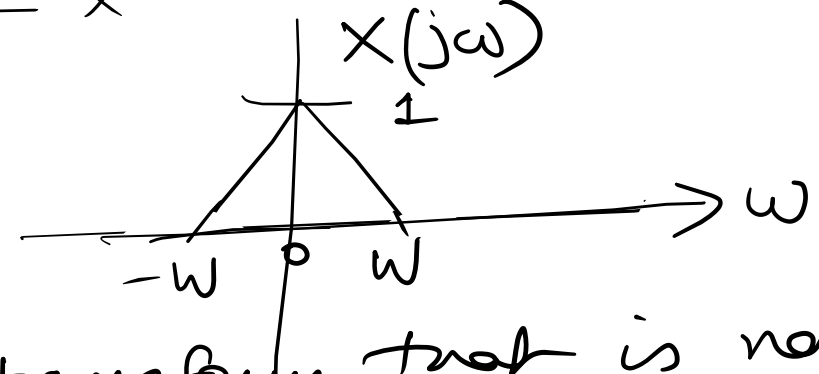
$$\frac{1}{2\pi}\left[\frac{2}{1+t^2} = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-|\omega|}\, e^{j\omega t}\, d\omega\right] \Rightarrow$$

$$\frac{1}{2\pi}\,\frac{2}{1+t^2} = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-|\omega|}e^{j\omega t}\,d\omega$$

$$\frac{1}{2\pi}\cdot\frac{2}{1+t^2} \Longleftrightarrow e^{-|\omega|}$$

$$\frac{2}{1+t^2} \Longleftrightarrow 2\pi e^{+|\omega|}$$

Ex / $x(t)$ is a C.T. aperiodic bandlimited signal



Signal $x(t)$ has a Fourier transform that is non-zero only over a finite band of frequencies & is zero otherwise // $X(j\omega)$ is bandlimited

$$p(t) = x(t)\cos\omega_0 t \qquad z(t) = \cos\omega_0 t$$

$$P(j\omega) = \frac{1}{2\pi}\{X(j\omega) * Z(j\omega)\}$$

$$p(t) = x(t)\,z(t)$$

$$P(j\omega) = \int_{-\infty}^{\infty}\left(x(t)\,z(t)\right)e^{-j\omega t}\,dt$$

$$= \int_{-\infty}^{\infty}\frac{z(t)}{2\pi}\left(\int_{-\infty}^{\infty}X(j\omega')\,e^{j\omega' t}\,d\omega'\right)e^{-j\omega t}\,dt$$

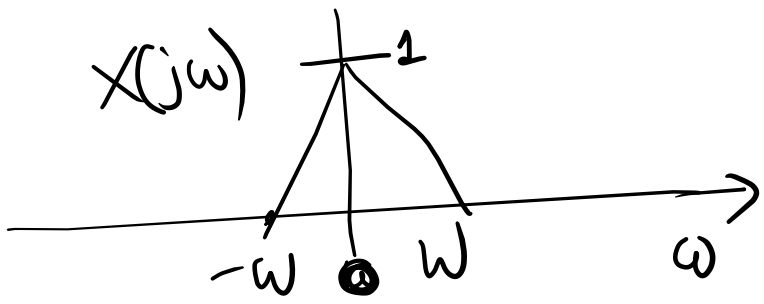
$$= \frac{1}{2\pi}\int_{-\infty}^{\infty}X(j\omega')\left(\int_{-\infty}^{\infty}z(t)\,e^{-j(\omega-\omega')t}\,dt\right)d\omega'$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty}X(j\omega')\,Z(j(\omega-\omega'))\,d\omega' = \frac{1}{2\pi}\left[X(j\omega) * Z(j\omega)\right]$$

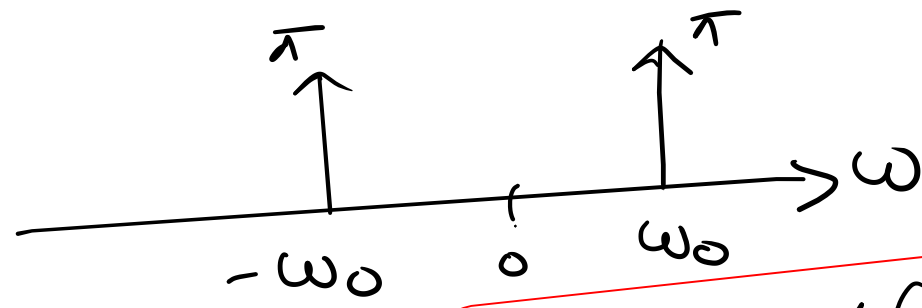
$$p(t) = x(t)\cos\omega_0 t = x(t)\,z(t)$$

$$P(j\omega) = \frac{1}{2\pi}\{X(j\omega) * Z(j\omega)\}$$

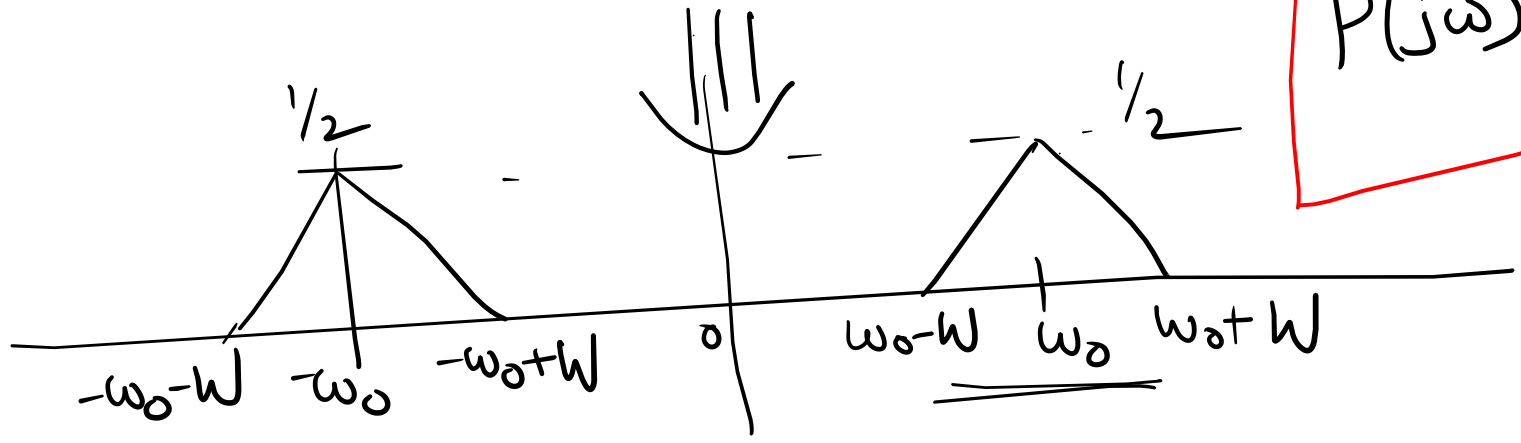
$$\cos\omega_0 t \rightleftharpoons \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)$$



$*$



Low pass signal



$$P(j\omega) = \frac{X(j(\omega-\omega_0))}{2} + \frac{X(j(\omega+\omega_0))}{2}$$

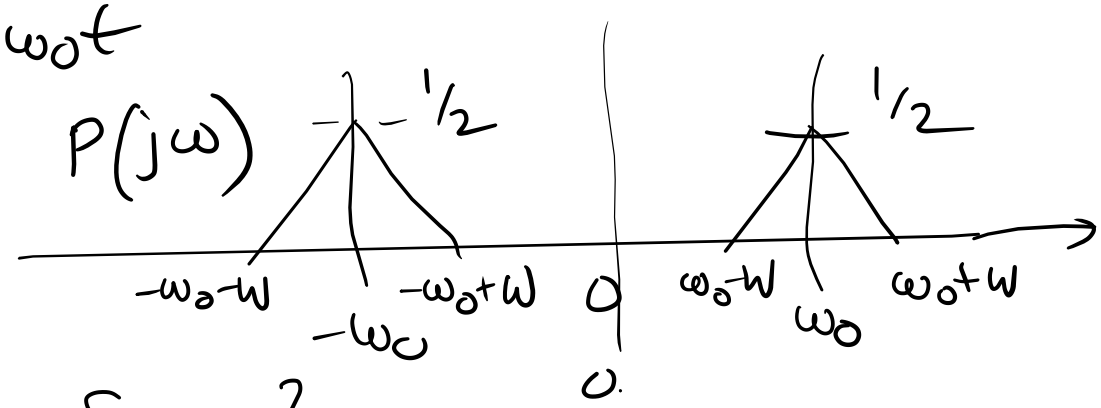
$$r(t) = p(t)\cos\omega_0 t$$

$$R(j\omega) = \frac{1}{2\pi}\{P(j\omega) * Z(j\omega)\}$$

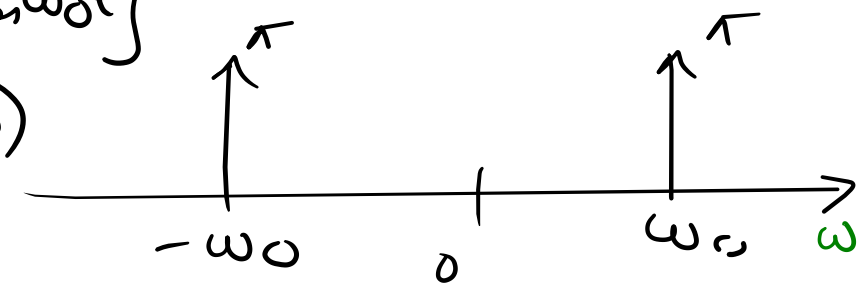
$$= \frac{1}{2\pi}\{P(j\omega) * \pi\delta(\omega-\omega_0)\}$$

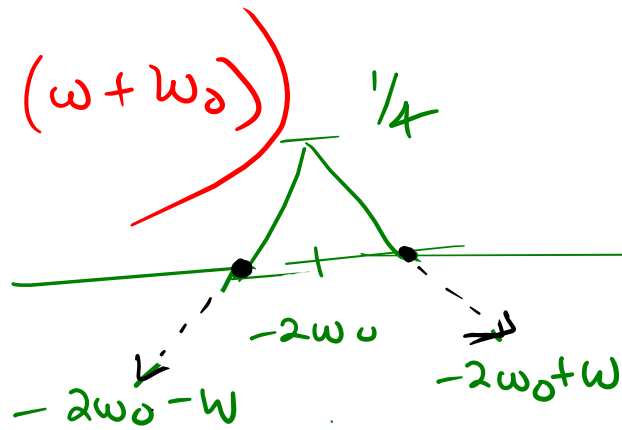
$$+ \frac{1}{2\pi}\{P(j\omega) * \pi\delta(\omega+\omega_0)\}$$

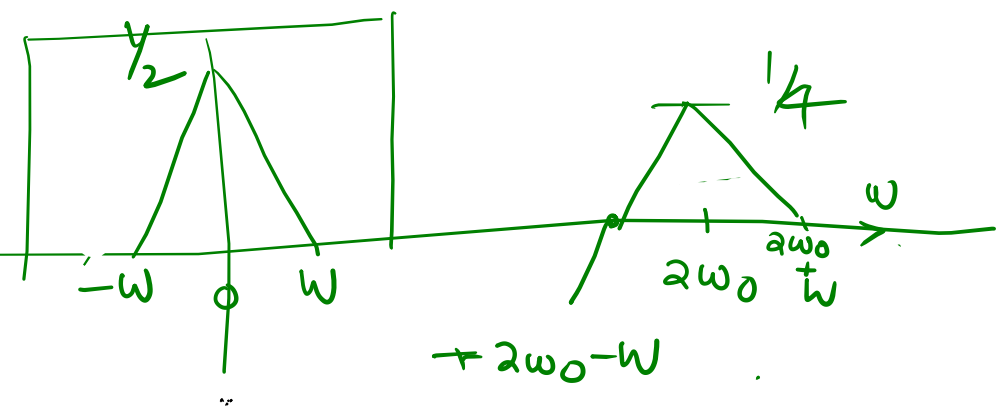
$$= \frac{1}{2}P(j(\omega-\omega_0)) + \frac{1}{2}P(j(\omega+\omega_0))$$



F.T.{cos ω₀t} = Z(jω)







Properties of F.T.

Linearity, time shifting, frequency shifting, multiplication (in time & frequency)

Scaling :

$$x(t) \rightleftharpoons X(j\omega)$$

$$y(t) = x(at) \rightleftharpoons ?$$

$$Y(j\omega) = \int_{-\infty}^{\infty} x(at)\, e^{-j\omega t}\, dt$$

$at = \gamma$

$a\,dt = d\gamma$

Case-1 $a > 0$

$$Y(j\omega) = \int_{-\infty}^{\infty} x(\gamma)\, e^{-j\frac{\omega\gamma}{a}} \cdot \frac{d\gamma}{a} = \frac{1}{a}\int_{-\infty}^{\infty} x(\gamma)\, e^{-j\left(\frac{\omega}{a}\right)\gamma}\, d\gamma$$

$$Y(j\omega) = \frac{1}{a}\, X\left(\frac{j\omega}{a}\right)$$

Case-2 $\quad a < 0$

$$Y(j\omega) = \int_{\infty}^{-\infty} x(at)\, e^{-j\omega t}\, dt$$

$at = \gamma$

$a\,dt = d\gamma$

$t \to \infty,\ \gamma \to -\infty$

$t \to -\infty,\ \gamma \to \infty$

$$= \int_{\infty}^{-\infty} x(\gamma)\, e^{-j\left(\frac{\omega}{a}\right)\gamma} \frac{d\gamma}{a}$$

$$= -\frac{1}{a} \int_{-\infty}^{\infty} x(\gamma)\, e^{-j\left(\frac{\omega}{a}\right)\gamma}\, d\gamma$$

$$= -\frac{1}{a} X\left(j\frac{\omega}{a}\right) = \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$$

In general, $\quad x(at) \rightleftharpoons \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$

Special case $a = -1$

$$x(at) \rightleftharpoons \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$$

$$x(-t) \rightleftharpoons X(-j\omega)$$

Reversal in time leads to reversal of F.T. on frequency axis.

Differentiation

$$x(t) \rightleftharpoons X(j\omega)$$

$$\frac{d}{dt}x(t) \rightleftharpoons j\omega X(j\omega)$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$\frac{d}{dt}x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, j\omega\, e^{j\omega t}\, d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left(j\omega X(j\omega)\right) e^{j\omega t}\, d\omega$$

Integration

$$x(t) \rightleftharpoons X(j\omega)$$

$$y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau \rightleftharpoons \frac{X(j\omega)}{j\omega} + \pi X(j0)\delta(\omega)$$

Show it!

$$= x(t) * u(t) \rightleftharpoons Y(j\omega) = X(j\omega)U(j\omega)$$

$$= X(j\omega)\left[\frac{1}{j\omega} + \pi\delta(\omega)\right]$$

Show $U(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$

$$= \frac{X(j\omega)}{j\omega} + \pi\delta(\omega)X(j\omega)$$

$$= \frac{X(j\omega)}{j\omega} + \pi X(j0)\delta(\omega)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

$$X(j0) = \int_{-\infty}^{\infty} x(t)\,dt$$

Differentiation in the freq. domain

$$x(t) \rightleftharpoons X(j\omega)$$

$$t\,x(t) \rightleftharpoons j\frac{d}{d\omega} X(j\omega)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$\frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} x(t)\,(-jt)\cdot e^{-j\omega t}\, dt$$

$$j\frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} (t\,x(t))\, e^{-j\omega t}\, dt$$

Parseval's theorem –

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

$|X(j\omega)|^2$ = energy density Spectrum

T.P.T. $\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t)\, x^*(t)\, dt$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$x^*(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X^*(j\omega)\, e^{-j\omega t}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} x(t)\left(\int_{-\infty}^{\infty} X^*(j\omega) e^{-j\omega t} d\omega\right) dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X^*(j\omega)\left(\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt\right) d\omega$$

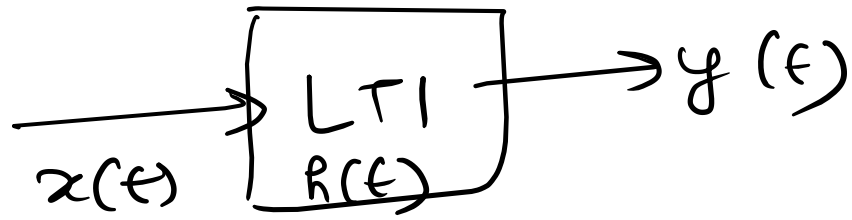
$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X^*(j\omega)\, X(j\omega)\, d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

Ex

Impulse response of an LTI system

$$x(t) = e^{-at}u(t) \qquad a>0$$

$$h(t) = e^{-bt}u(t) \qquad b>0$$



$$y(t) = x(t) * h(t)$$

$$Y(j\omega) = X(j\omega)H(j\omega)$$

$$= \frac{1}{a+j\omega} \cdot \frac{1}{b+j\omega}$$

$$= \frac{1}{b-a}\left\{\frac{1}{a+j\omega} - \frac{1}{b+j\omega}\right\}$$

$$y(t) = IFT\left(Y(j\omega)\right)$$

$$= \frac{1}{b-a}\left\{e^{-at} - e^{-bt}\right\}u(t)$$

Let $a = b$

$$x(t) = h(t) = e^{-at} u(t)$$

$$Y(j\omega) = X(j\omega) H(j\omega)$$

$$Y(j\omega) = \left(\frac{1}{a+j\omega}\right)^2$$

$$\boxed{y(t) = t e^{-at} u(t)}$$

$a > 0$

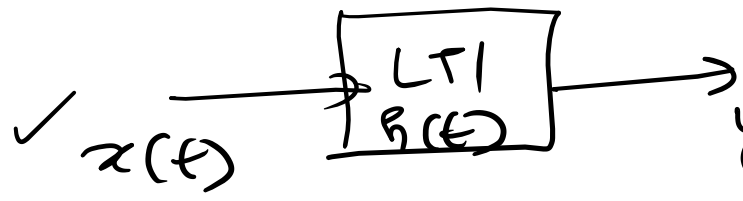
$$R(j\omega) = \frac{1}{a+j\omega}$$

$$\frac{d}{d\omega} R(j\omega) = \frac{-j}{(a+j\omega)^2}$$

$$j \frac{d}{d\omega} R(j\omega) = \left(\frac{1}{a+j\omega}\right)^2$$

$$\Rightarrow t r(t)$$

Ex Let $x(t) = (e^{-t} + e^{-3t})u(t)$ & $y(t) = (2e^{-t} - 2e^{-4t})u(t)$



Given $x(t)$ & $y(t)$, find $h(t)$?

Problem of System identification

Sol

$$Y(j\omega) = X(j\omega)\,H(j\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

$$X(j\omega) = \frac{1}{1+j\omega} + \frac{1}{3+j\omega}$$

$$Y(j\omega) = \frac{2}{1+j\omega} - \frac{2}{4+j\omega}$$

$$= \frac{\frac{2}{1+j\omega} - \frac{2}{4+j\omega}}{\frac{1}{1+j\omega} + \frac{1}{3+j\omega}}$$

$$= \frac{(1+j\omega)(3+j\omega)}{(1+j\omega)(4+j\omega)} \times \frac{6}{4+2j\omega}$$

$$= \frac{6(3+j\omega)}{(4+j\omega)(2+j\omega)}$$

$$H(j\omega) = \frac{3(3+j\omega)}{(4+j\omega)(2+j\omega)}$$

$$= \frac{A}{4+j\omega} + \frac{B}{2+j\omega}$$

$A = 3/2$

$B = 3/2$

$$H(j\omega) = \frac{3/2}{4+j\omega} + \frac{3/2}{2+j\omega}$$

$$h(t) = \frac{3}{2} e^{-4t} u(t) + \frac{3}{2} e^{-2t} u(t)$$

Ex Consider a stable & causal LTI system defined by the below eqns —

$$\frac{d}{dt} y(t) + a y(t) = x(t) \quad ; \quad a > 0$$

what is $h(t)$?

Take F.T. on both the sides —

$$j\omega Y(j\omega) + a Y(j\omega) = X(\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{1}{a+j\omega}$$

$h(t) = \mathcal{F}.\mathcal{T}.^{-1}\left\{\frac{1}{a+j\omega}\right\}$

$= e^{-at} u(t)$

$a > 0$

Ex-2

A causal LTI system is given by —

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = \frac{d}{dt}x(t) + 2x(t)$$

Find $h(t)$?

$$(j\omega)^2 Y(j\omega) + 4j\omega Y(j\omega) + 3Y(j\omega) = j\omega X(j\omega) + 2X(j\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{j\omega + 2}{(j\omega)^2 + 4j\omega + 3}$$

$$= \frac{j\omega + 2}{(j\omega + 1)(j\omega + 3)} = \frac{A}{j\omega + 1} + \frac{B}{j\omega + 3}$$

$A = \frac{1}{2}$

$B = \frac{1}{2}$

$$h(t) = \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{-3t}u(t)$$

Given a causal LTI system characterized by

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

$$H(j\omega) = \frac{\sum_{k=0}^{M} b_k (j\omega)^k}{\sum_{k=0}^{N} a_k (j\omega)^k} = \frac{Y(j\omega)}{X(j\omega)}$$

$=$ ratio of polynomials in $(j\omega)$

$=$ Rational fn. in $(j\omega)$

$=$